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ENG 1102

Herman Isayenka, Yui Him Au, Berke Ucuncuoglu, Matteo Condello, Mohammad Tawil

OT2T4_Phase2_Report

1. Problem Restatement and Linkage to Phase 1

In phase 1, the problem identified was that seniors aged 65 and older in Ontario struggle to keep their driveways and outdoor paths clear during the winter. Snow and ice make shovelling physically demanding and dangerous, especially in cold, slippery conditions. Existing snow removal methods, such as manual shovelling or relying on city services, are often slow, physically difficult, or unreliable. This leaves many seniors at risk of injury and makes it harder for them to access their driveways safely.

Insights gathered from in-person interviews and Google Form surveys helped guide the direction of our solution. Many interviewees explained that as they age, physical tasks, such as shovelling snow, become harder to do and increase their risk of injury. The majority of participants of the survey wrote that they have experienced a slip or fall on their driveway while shoveling. These insights suggest that potential solutions should focus on reducing or eliminating the physical effort required and improving safety while clearing snow from driveways.

Based on these insights, we developed a set of criteria for success that we used to evaluate potential solutions. First, the solution should significantly improve safety by reducing the risk of slips and falls caused by snow and ice. Second, it should minimize the physical effort required for winter driveway maintenance. Finally, the solution should be reliable and effective during typical winter conditions, ensuring that users can consistently keep their outdoor paths clear.

2. Divergent Ideation & Exploration

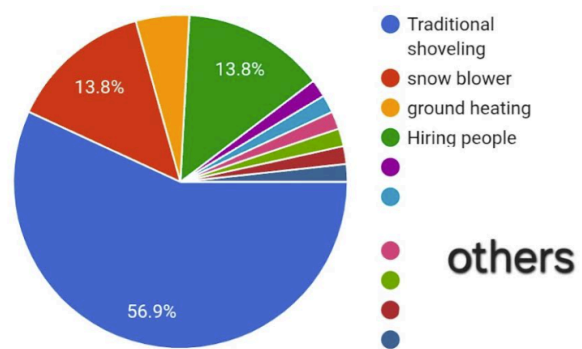
2.1 Research of Existing Solutions:

After we did some surveys and some research online, excluding parts of the percentages for the people who live in apartments and condos.

We discovered 5 main methods

What equipment or methods do you use when comes to cleaning up the snow:

58 responses



- **Traditional shovelling:** The most traditional method, with 56.9% of our users

still using this, and being the highest on the chart, it is highly accessible, costing approximately \$50 with a lifespan of several years, and requires minimal storage space. However, it's limited to the user's health condition as well as age. It is a very physically demanding task with the risk of icy sidewalk, and physical injury with the repetitive motion of bending, lifting, and twisting puts immense pressure on your lower back and shoulders, adding to the potential endangerment to your physical condition.

- **Snowblower:** Snowblower is the 2nd most popular used solution with 13.8%, the main method is to have a spinning auger to push snow into a discharge chute to be thrown to the side and out of the way. It reduces the physical work required by simply pushing or even driving around the snow blower, depending on the type. The limitations of the method are due to pricing and storage, with the average price range ranging between \$300 to \$1200; the quality itself

would vary depending on the price, which could affect the performance of the snowblower.

Storage can be an issue for some people with smaller houses, as it can take up space.

- **Hiring a worker:** Due to old age or personal reasons, sometimes people are unable to do their own shovelling in the driveway, so they choose to hire people to do the snow service. It comes in two types, with some being professional and the other is hiring people from around the neighbourhood. The charging fee usually ranges between \$30 to \$100 per service. But this method has some uncertainty, especially when the weather gets extreme. Will the company or the person you pay be willing to work in these conditions? Also, in an emergency, they might not be able to respond in time to help you clean the driveway.

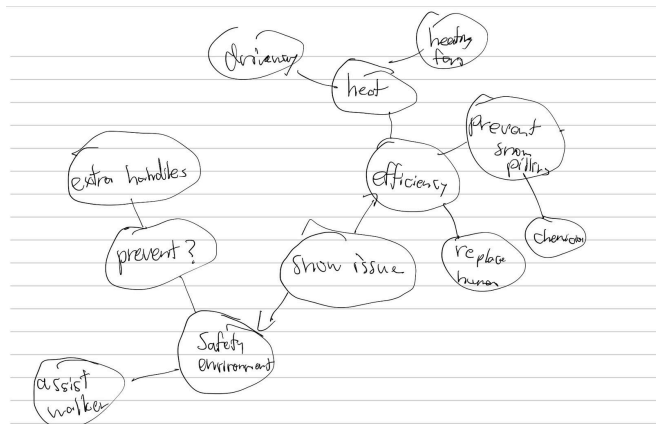
- **Heating Driveway:** Heating the driveway is a method of putting heating elements beneath the surface to keep areas clear of ice and snow. It has two types of heat systems: electric and hydronic. Electric systems use conductor resistance heat cable placed roughly 2 inches below the surface, while the Hydronic system uses a water pump and boiler to heat the ground. It is perhaps the most convenient system among the solutions as it is fully automated, and it requires zero manual labour when it comes to snow. The biggest takeaway from this method is the budget, since installation costs about \$6.52 to \$8.69 per square foot and \$3.55 per hour of operation.

- **Chemical salt:** While not captured in the survey, the use of de-icing agents is a common supplemental method. These products are highly effective, capable of melting ice in temperatures as low as -31.67°C . Users typically apply these to entryways and steps overnight for quick

clearing. However, frequent application presents significant drawbacks; it can accelerate the degradation of concrete, particularly in newer or unsealed surfaces, leading to costly long-term repairs. With every bag costing around \$30 and you'd prefer not to overuse it before it damages the driveway, people often use it only for a simple layer.

2.2 Ideation Process & Breadth of Idea Generation

During this phase of the design process, we focused on making sure that there are many different solutions to our problem and making sure that we select



the best option. We started off with a session where we all shared our ideas that would help the elderly homeowners to safely clear ice and snow from their driveways with minimal physical effort as shown in the graph the people would prefer a safer and more efficient solution.

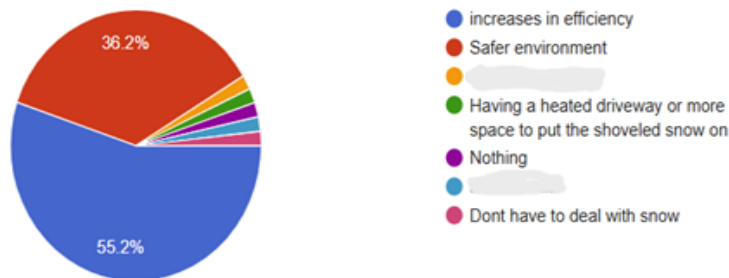
Brainstorming allowed us to quickly list traditional methods like snowblowers. Applying "Eliminate" from SCAMPER pushed us to think of solutions that remove the user from the driveway entirely, leading to the heating system. These results that were shared have been recorded in order to make sure we used creative thinking. This process helped us consider multiple options, like mechanical devices, robotic systems, and chemical modifications. For example, one of the ideas was a chemical salt that can be highly effective and requires minimal physical effort. Another example is the heated driveway that can help melt snow and ice. In the

end we came up with 7 unique solutions. We evaluated them using the criteria for success, which helped us to narrow down and choose the option we think best fixes the problem. You can also see that above we conducted a survey that helped us to learn and narrow down the solution that best fits the problem. We can see that in the survey from the previous, 56.9% of users are still using the traditional shovelling method. Although this method is likely the most cost-friendly, it is very physically demanding, and many elderly people can be severely injured using this method. Overall, the divergent ideation process helped us to explore a lot of different ideas and break down the final concept that is presented in the upcoming sections.

What improvements would help you most during heavy snow periods?

 [Copy chart](#)

58 responses

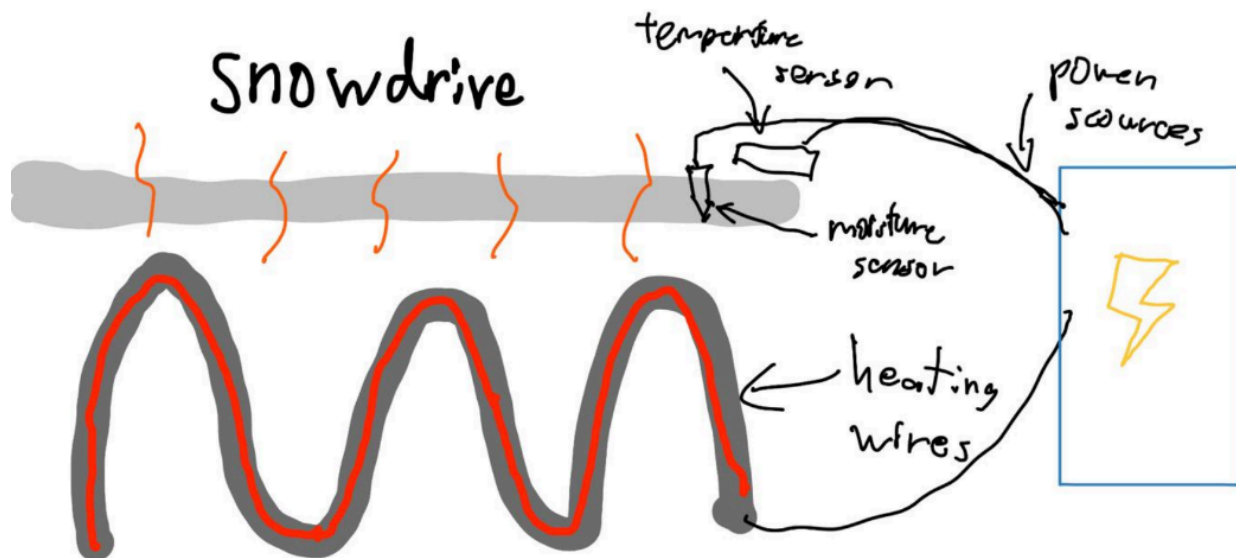
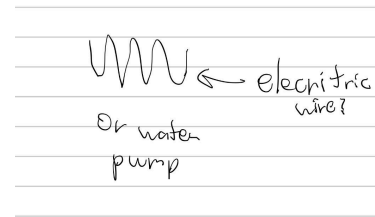


3. Three Feasible Concept Developments

To tackle our concepts, we developed 3 solutions, with each having distinct functions and focus on a different area, while all aiming to make the situation a safer and more efficient environment:

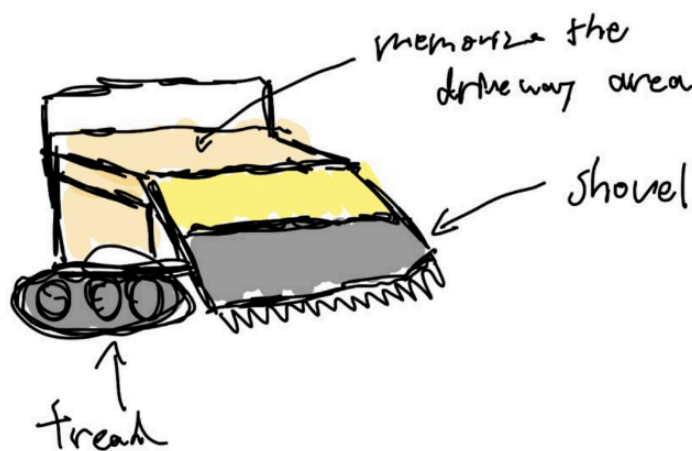
Concept A: Snowdrive

Among the solutions identified in our research, we found that a heated driveway is the most suitable and the closest to perfection. The same goes for our Snowdrive, with the idea of installing a temperature and moisture sensor at the front of your house. Once the sensor detects the temperature is below a certain degree, as well as the moisture sensor senses the ground is unusually wet, both of them will send a signal to the heating material beneath the driveway, heating the ground instantly and preventing any snow from piling up. Making sure you can walk in and out of your house safely without any effort to shovel or even take any action against the snow outdoors.



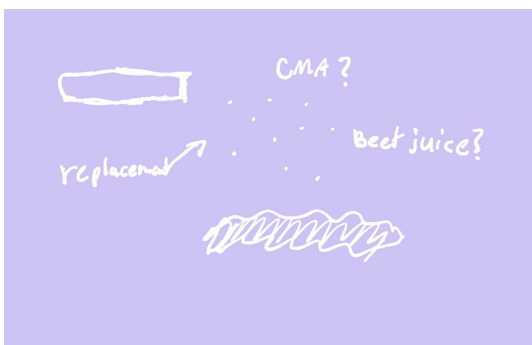
Concept B: Frosty

Frosty is a snow-shovelling robot designed to help replace physical work. It is a robot installed with treadmill tires helping it to move around the snowy driveway without any issues, and a shovel in front of it. It is designed to remember the pattern of your driveway, and once deployed it would automatically start shoveling any objects in front of it, as it goes around the driveway, until the work is done, and it would return to its place.



Concept C: Breaking Ice

The third concept, Breaking Ice shifts the focus from mechanical labour to chemical efficiency. Our goal here is to discover an alternative to traditional road salt that can significantly accelerate the melting process. This is the most straightforward solution to implement because it requires no complex machinery or expensive driveway construction. Alternatives such as beet juice, or



CMA (Calcium Magnesium Acetate) can be used as an alternative as they can help prevent the ice from piling before it even becomes an issue.

4. Concept Selection and Justification

4.1

Using the data from our matrix chart:

- Desirability has a weight of 30%
- Feasibility has a weight of 30%
- Viability has a weight of 20%
- Responsibility has a weight of 20%
- Creativity has a weight of 0%

Feasibility was identified to be one of the most important dimensions because a key requirement of this project is to propose a solution that can realistically be implemented using existing technology and infrastructure. Given the constraints of time, cost, and available resources, it is important that the proposed solution can be built without highly specialized equipment or unrealistic modifications.

Desirability was the other most important design dimension because the solution must effectively address the needs of homeowners dealing with snow and ice accumulation on driveways. From discussions with potential users convenience and reliability were identified as the most important factors.

Creativity was weighted as the least important design dimension because the primary goal of the project is to develop a reliable and effective solution rather than a highly novel one. While the innovation is valuable the users will prioritize the consistent solutions rather than the unique ones.

4.2 Customized Scoring Rubrics

Score Scale: 0 = Failure, 1 = Very poor, 2 = Poor, 3 = Average, 4 = Good, 5 = Excellent

Desireability: Does this solution address user needs for removing snow and ice?

- 0: The design does not remove snow or ice and may worsen conditions
- 1: Provides little improvement compared to existing solutions
- 2: Attempts to reduce snow or ice but has usability issues for regular use
- 3: Addresses basic snow removal needs but requires effort or additional steps
- 4: Meets user needs by effectively with only minor limitations
- 5: Completely satisfies user needs by always keeping driveways clear

Feasibility: *Can the solution realistically be implemented?*

- 0: The concept cannot realistically be built

- 1: Extremely difficult to implement due to technological limitations
- 2: Requires specialized equipment/ materials that is difficult to access
- 3: Moderately complex solution that may require complex components
- 4: Mostly feasible with existing technology
- 5: Highly feasible using proven technology and existing systems

Viability: Can the solution function effectively in winter over time?

- 0: The concept is impractical due to high cost and poor reliability
- 1: The solution works poorly in winter conditions
- 2: The solution partially works but contains major limitations such as cost or maintenance requirements
- 3: The solution works under certain conditions but not all
- 4: The solution works effectively with minor long term limitations
- 5: The solution is reliable cost effective and capable of functioning throughout winter conditions

Responsibility: Is the solution safe and environmentally responsible?

- 0: Creates significant damage to environment as well as safety risks
- 1: Shows little consideration for environment and safety of users
- 2: Displays environmental and safety concerns that are not fully addressed
- 3: Demonstrates moderate consideration of environmental and safety impacts
- 4: Environmentally responsible and safe with users with minor concerns
- 5: Strongly prioritizes safety and environmental responsibility

Creativity: Does the solution introduce innovation?

- 0: Direct copy of existing methods with no innovation
- 1: very similar to existing solutions with minimal improvement
- 2: Minor improvements over traditional approaches
- 3: Introduces new features to current solutions
- 4: Clearly improves upon existing snow removal solutions
- 5: Highly innovative solution that reshapes how the same problem can be fixed

4.3 Matrix Table

	Desireability	Feasibility	Viability	Responsibility	Creativity	Total
Desireability	x	4	4	4	5	17
Feasibility	1	x	4	4	5	14
Viability	1	1	x	3	4	9
Responsibility	1	1	2	x	4	8
Creativity	0	0	1	1	x	2

5. Interpretation and forward plans

The final concept selected is Concept A, “snowdrive,” a heated driveway system designed to automatically melt snow and ice. This concept was chosen because it best addresses the needs of users, particularly elderly homeowners, by reducing the physical effort required to clear snow and helping improve safety during winter conditions. By automatically melting snow and ice, the system can help prevent slips and reduce the risks associated with manual snow removal.

However, a key weakness of this concept is the cost and complexity of installation, since the heating system must be placed beneath the driveway surface. This may require significant construction work and could make the system less affordable for some homeowners.

In phase 3, we plan to test key assumptions by developing and evaluating a small-scale heated surface prototype to observe how effectively it melts snow and ice under winter-like conditions. We will also assess energy efficiency and explore potential ways to simplify installation and reduce overall costs, which could improve the practicality and accessibility of the system for the homeowners.

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Contribution Table

Name	Contribution	Type of task	Estimated Time Spent
Herman	Part 4.1,4.2,4.3	Matrix calculation, Project Management, communication	1h 45 min
Yui Him	Part 2.1, 3, Sketches and references	Research, Design, Communication	2 hrs
Tawil	Part 5	Communication, Teamwork, Organizing	2 h
Berke	Part 1	Communication, Teamwork	2h
Matteo	Part 2.2	Ideation Process	1h 45min